Paleomagnetism of Keweenawan Intrusive Rocks, Minnesota

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A paleomagnetic pole position calculated for 131 sites in normally (N) magnetized Keweenawan intrusive rock from Minnesota lies at 34.5°N, 168.5°W. Gabbro and diabase of reverse (R) polarity also are known but only from a limited area in northern Cook County, near the Canadian border. R-polarity sites yield a pole position of 42.5°N, 156.5°W and probably are older than the N-polarity sites. Magnetic intensities in these rocks vary by 4 orders of magnitude, in keeping with other strong differences in lithology, which together reflect active igneous differentiation. Dispersion of magnetic directions in the intrusive rocks indicates that the ratio of nondipole to dipole field-intensity has not changed markedly since Keweenawan time. It is suggested that, on the basis of current evidence, apparent polar wandering relative to North America is best described by a rather vaguely defined southwestwardly trending path several tens of degrees in width. Under favorable circumstances positions within the path of polar wandering can be used for geological correlation. Several episodes of Precambrian basaltic volcanism in North America are used to illustrate long-distance correlation by paleomagnetic techniques.

This report is a result of a general investigation by the U.S. Geological Survey of the paleomagnetism and magnetic intensities of Keweenawan igneous rocks exposed in the Lake Superior syncline south of the Canadian border. Purposes of the larger investigation are: (1) to investigate the fine-structure of polar wandering relative to northern North America during the late Precambrian; (2) to search for geomagnetic reversals for use as time horizons in stratigraphic studies; (3) to investigate the strength of geomagnetic secular variation during the lafe Precambrian; (4) to use directions of remanent magnetization to test for the existance of large-scale rotations of crustal blocks in the Lake Superior area. Magnetic intensity data of value in the interpretation of aeromagnetic maps also are being collected. Reports issued to date have described results for the Beaver Bay Complex, Minnesota [*Beck and Lindsley*, 1969], the gabbroic rocks near Mellen, Wisconsin [*Books et al.*, 1966] and the lowermost Keweenawan lava flows in Michigan, Wisconsin and Minnesota [*Books*, 1968]. The present report summarizes paleomagnetic data for several large bodies of Keweenawan intrusive rock exposed in Minnesota. Similarity of paleomagnetic pole position, radiometric age dates, and geological setting indicate that these intrusive rock units formed as part of a single igneous event. The purpose of this report is to derive a reliable paleomagnetic pole position for these intrusive rock units and to compare this pole position with other North American pole positions of roughly similar age.

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Earlier paleomagnetic investigations of Keweenawan intrusive rocks in Minnesota are limited to the reconnaissance of the Duluth gabbro near Duluth by *DuBois* [1962], to work on the Endion sill by *Helsley* [1965], and to the recent investigation of the Beaver Bay Complex by *Beck and Lindsley* [1969]. Results of these investigations are summarized in Table 1.

GEOLOGICAL SUMMARY

Intrusive rocks that accompanied the great outpouring of volcanic rocks in the Lake Superior area during Keweenawan time are now exposed mainly on the northwest flank of the Lake Superior syncline [*Halls*, 1966; *White*, 1966*a*, *b*]. Gabbro, diabase, and related mafic lithologies greatly predominate among Keweenawan intrusive rocks; a few rocks of granitic composition occur locally, but in minor amounts. Potassium-argon ages for these intrusive rocks

TABLE 1. Summary of Earlier Paleomagnetic Investigation of Intrusive Rocks of Keweenawan Age from Northern Minnesota

Unit	No. Sites	λ'	ϕ'	Reference	Comments
Duluth gabbro complex	9	23.5°N	179.5°E	*DuBois* [1962]	a.c. demagnetization to 200 oe. Apparently no tectonic correction applied.
Endion sill		27.5°N	178.0°W	*Helsley* [1965]	Results cited as 'near' the location shown. Tectonic correction of unstated amount applied.
Beaver Bay complex	29	27.5°N	170.5°W	*Beck and Lindsley* [1969]	a.c. demagnetization to 300 oe applied. Results quoted are for 75 oe level. No tectonic correction necessary (igneous lamination horizontal).

λ' is latitude and ϕ' is longitude of the paleomagnetic pole.

fall in the range 0.95 to 1.2 b.y. [*Goldich et al.*, 1961]; ages of 1.1 b.y. have been reported by *Faure et al.* [1969], using Rb/Sr methods, and by *Silver and Green* [1963] using zircons. Inclined layering and lamination indicate that minor post-consolidation warping has affected the Keweenawan intrusive rocks in Minnesota. For the purposes of this study dips of 20°E, 20°S, and horizontal have been adopted for the Duluth area, the Cook County sampling area, and the Beaver Bay area, respectively. Locations of major rock units in Minnesota discussed in this report are shown in Figure 1.

Experimental Methods

Samples were collected by means of a water-cooled, gasoline-powered diamond drill; orientations were applied by means of solar compass or magnetic compass carefully corrected for local declination anomalies. The number of samples collected at most sites ranged from 1 to 7 and in a few instances was as high as 19. This extreme range resulted from the needs of magnetic properties studies [*Beck*, 1969], which made it important to obtain examples of the full range of lithology at each site. Details of sampling techniques and site localities are reported in *Beck* [1969].

Directions of remanent magnetization of most specimens were measured on a version of the spinner magnetometer described by *Doell and Cox* [1965]. A few weakly magnetized samples were measured on a more sensitive spinner magnetometer designed by *Phillips and Kuckes* [1967]. In their range of overlap these two instruments gave entirely consistent results.

All specimens were magnetically cleaned in peak alternating fields (af) of 75, 150, and 300 oe, using a standard three-axis tumbler [*Beck*, 1969]. Specimens that continued to yield strongly divergent directions of magnetization after cleaning at 300 oe were also subjected to demagnetization at 600 oe. Optimum field for magnetic cleaning of each site was chosen on the basis of minimum within-site dispersion for sites at which $N \geq 2$ [*Irving et al.*, 1961]. Sites consisting of a single sample were regarded as magnetically clean only if the sample direction reached a 'stable end point' [*McElhinny and Gough*, 1963]. Figure 2 is a histogram of optimum fields for af magnetic cleaning of Keweenawan intrusive rocks.

Some sites failed to form stable groupings even after demagnetization in fields as high as 600 oe. Use of 'average directions' from such sites in calculations of mean directions for lithological units would tend to introduce dispersion not related to any directional properties of the paleogeomagnetic field. For this reason it was necessary to adopt rejection criteria for the elimination of unstable sites. The criteria adopted were as follows. (1) For sites with $N > 2$ the tables of $R_0(N)$ in *Irving* [1964, p. 63] were used. $R_0(N)$ is a length exceeded with a probability of 5% by the vector sum of N randomly oriented unit vectors.

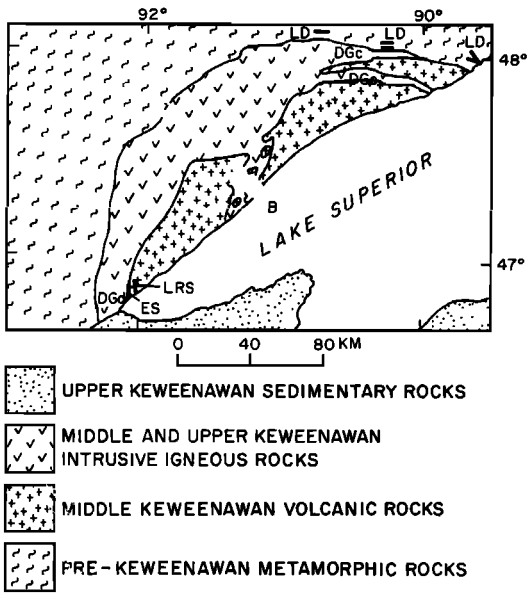


Fig. 1. Geological sketch map of the southwestern Lake Superior area, showing sampling localities. Modified from *White* [1966a]. DGd, Duluth gabbro complex at Duluth (including anorthositic gabbro, layered series, Bardons Peak gabbro of *Taylor* [1956], ferrogranodiorite, and granophyre); ES, Endion sill; LRS, Lester River sill; BB, Beaver Bay complex of *Beck and Lindsley* [1969]; DGc, Duluth gabbro complex in Cook County; LD, Logan diabase.

If $R \leq R_0$, the site was discarded. (2) If $N = 2$, sites were rejected if the angle between the two sample directions exceeded 25° at every demagnetization level. The selection of 25° was entirely arbitrary. (3) All sites for which $N = 1$ were rejected unless a definite 'stable end point' was reached. For this study samples were considered to have reached a 'stable end point' if they changed direction by less than 5° between successive demagnetization levels. Of 144 sites, 13 were rejected as unstable on the basis of these criteria. Similar rejection criteria were adopted by *Robertson* [1969] for his study of the Muskox intrusion and related rocks. In general, the magnetic stability of Minnesota Keweenawan intrusive rocks is moderately high. Dispersion for most sites decreased slightly or was unaffected by magnetic cleaning. Fisher's precision parameter k [*Fisher*, 1953] for $N > 2$ sites has an approximate log-normal distribution with range 8 to 7435 and geometrical mean near 80. Highest stability is found in the Endion and Lester River sills. Lowest stability is found in the granophyre.

Results

Sites described in this section are located in two general areas: Duluth and vicinity in the south and Cook County in the north (see Figure 1). Data from sites in the Beaver Bay area, located approximately midway between the Duluth and Cook County sampling areas, have reported earlier [*Beck and Lindsley*, 1969]. Beaver Bay sites are included in the analysis presented later in this paper. Details of site localities and statistics are given in *Beck* [1969, Appendix].

The Duluth gabbro complex was sampled at 43 sites distributed over 60 km² in the Duluth metropolitan area. This complex gabbro body is regarded by *Taylor* [1964] as composed of the following distinct lithological units, listed in probable order of decreasing age:

1. Anorthositic gabbro: Massive, coarse-grained gabbro containing 75–90% calcic plagioclase. Contains many anorthosite inclusions.

2. Layered series: Medium-grained gabbro characterized by strong local development of rhythmic layering and igneous lamination. Cryptic layering (progressive upward change in mineral composition) is absent, in contrast to similar stratiform intrusions elsewhere [*Wager and Deer*, 1939; *Hess*, 1960; *Ford and Boyd*, 1968]. *Taylor* [1964] speculates that development of cryptic layering may have been impeded by renewal of the layered series magma one or more times during crystallization.

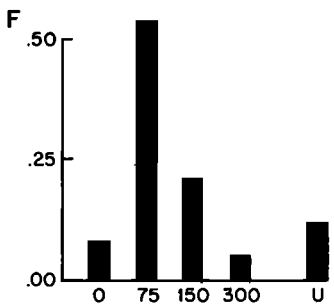


Fig. 2. Histogram of optimum fields for alternating field magnetic 'cleaning.' F denotes fraction of total sites. Numbers on abscissa indicate peak value of alternating field, in oersteds. U shows unstable sites.

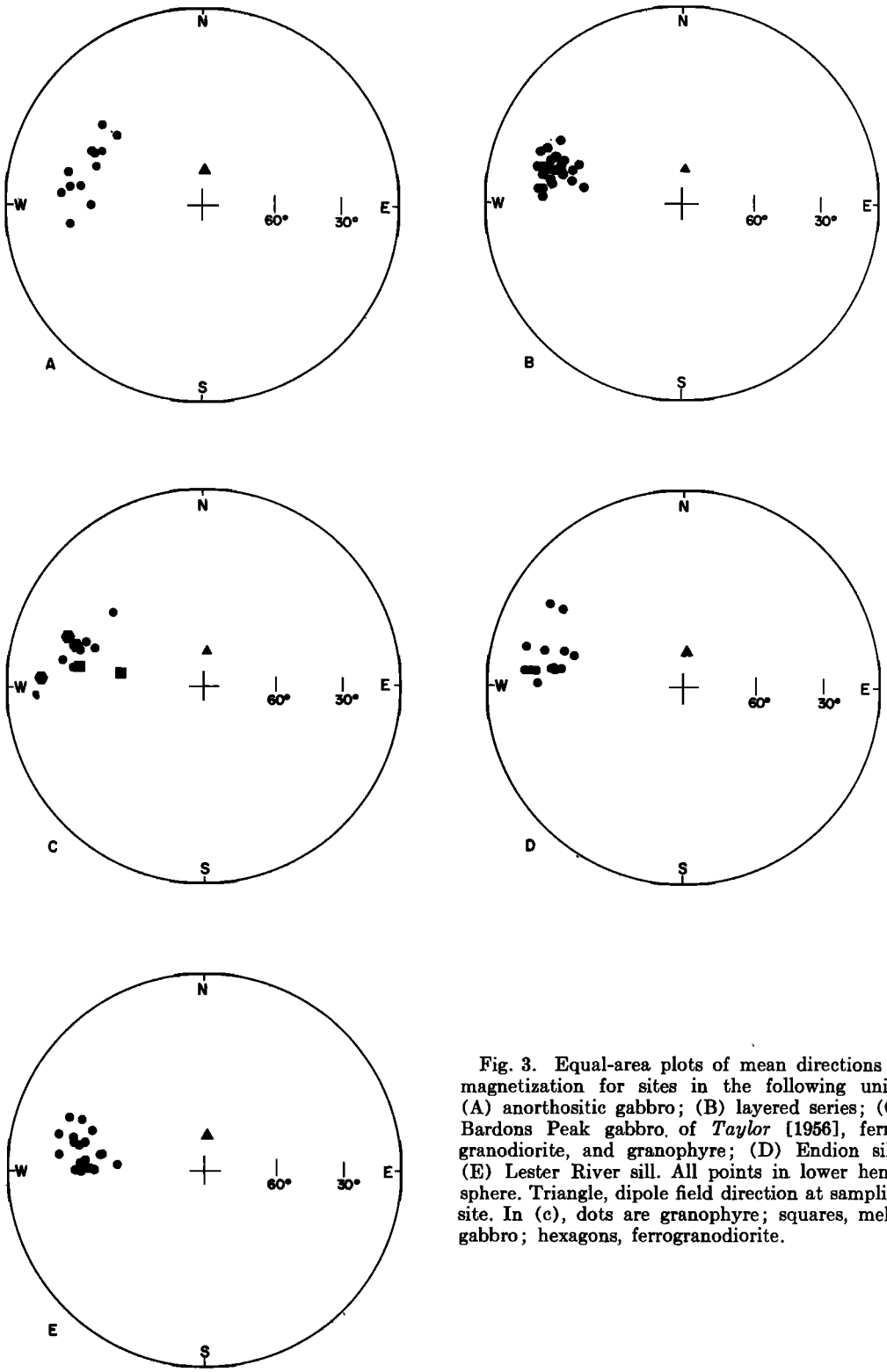


Fig. 3. Equal-area plots of mean directions of magnetization for sites in the following units. (A) anorthositic gabbro; (B) layered series; (C) Bardons Peak gabbro of *Taylor* [1956], ferrogranodiorite, and granophyre; (D) Endion sill; (E) Lester River sill. All points in lower hemisphere. Triangle, dipole field direction at sampling site. In (c), dots are granophyre; squares, melagabbro; hexagons, ferrogranodiorite.

TABLE 2. Direction of Remanent Magnetization and Paleomagnetic Pole Positions for Rock Units Near Duluth

Unit	No. Sites	D	I	k	α_{95}, deg	λ'	ϕ'	δP_{95}, deg	δm_{95}, deg
Anorthositic gabbro	12	294.5	+56.5	39.8	7.0	42.0°N	169.0°W	7.5	10.0
Layered series	24	290.0	+53.0	116.2	3.0	36.5°N	170.0°W	2.5	4.0
Endion sill	14	286.0	+49.0	53.3	5.5	31.5°N	171.0°W	5.0	7.5
Lester River sill	20	283.0	+57.5	78.9	3.5	35.0°N	161.0°	4.0	5.0

D, mean declination; I, inclination; k, precision parameter; α_{95}, radius of cone of 95% confidence [*Fisher*, 1953]; λ', latitude of paleomagnetic pole; ϕ', longitude of paleomagnetic pole; δP_{95} and δm_{95}, semi-axes of the oval of 95% confidence. For further discussion of these statistics see *Irving* [1964].

3. Bardons Peak gabbro of *Taylor* [1956]: Small body of coarse-grained olivine gabbro and peridotite, intruded into layered series rocks. Bardons Peak rocks may represent a briefly mobilized early mafic phase of the layered series.

4. Ferrogranodiorite, granophyre: Small units apparently formed as late differentiation products of gabbroic magma. Ferrogranodiorite is a medium-grained massive iron-rich pyroxene granodiorite; granophyre is a silica-rich rock composed largely of finely intergrown quartz and alkali feldspar.

Two bodies outside the Duluth gabbro complex also were sampled in the Duluth area. These are the Endion and Lester River diabase sills [*Schwartz and Sandberg*, 1940], located in Keweenawan lava flows stratigraphically just above the Duluth gabbro complex. These sills undoubtedly are closely related in mode and time of origin to the gabbro complex rocks. The Endion and Lester River sills are fine- to medium-grained tabular bodies of diabase and granophyre. Their thicknesses where exposed along the shore of Lake Superior are: Endion, 400 meters; Lester River, 270 meters. In both cases their complete thicknesses were sampled.

Mean directions of magnetization for sites from the Duluth area are plotted in Figure 3. Too few melagabbro, ferrogranodiorite and granophyre sites are available to justify separate plots or pole-position calculations; the 11 sites from these late units are shown together in Figure 3C. The directions shown in Figure 3 have not been corrected for postmagnetization tilting, but the paleomagnetic directions for these rocks summarized in Table 2 have been corrected for a dip of 20°E.

Units sampled in Cook County consist of the Duluth gabbro complex and the Logan diabase intrusions [*Grout and Schwartz*, 1933; *Grout et al.*, 1959]. Sampling sites were distributed over a wide area but were dominantly located along several roads that traversed the entire gabbro outcrop area. Rocks mapped as Duluth gabbro complex in Cook County show as much lithological variability as do Duluth gabbro complex rocks near Duluth, and crosscutting relationships between distinct gabbro types are found in many exposures. From this evidence it appears probable that detailed mapping will show that the Duluth gabbro complex of Cook County is a multiple intrusion similar to Taylor's Duluth gabbro complex [*Taylor*, 1964] and to Gehman's Beaver Bay Complex [*Gehman*, 1957; *Beck and Lindsley*, 1969] farther south. In the current study, Duluth rocks in Cook County are treated as a single unit, subdivided only on the basis of magnetic polarity. Presence of the reversely magnetized gabbro at the base of the intrusion suggests that these rocks are older than the normally magnetized gabbro [*Beck*, 1969].

Logan diabase sheets are common as intrusions into the Rove slate and are well exposed near Thunder Bay in Ontario and south to the U.S. border. Similar intrusions are found immediately north of the Duluth gabbro complex exposures in northermost Cook County. Sills, dikes, and irregular intrusions are present. Tabular bodies range in thickness from several meters to several hundred meters. Normal N and reverse R magnetic polarities both are found in Logan rocks, with the N group being the younger [*DuBois*, 1962].

Mean directions of magnetization for sites

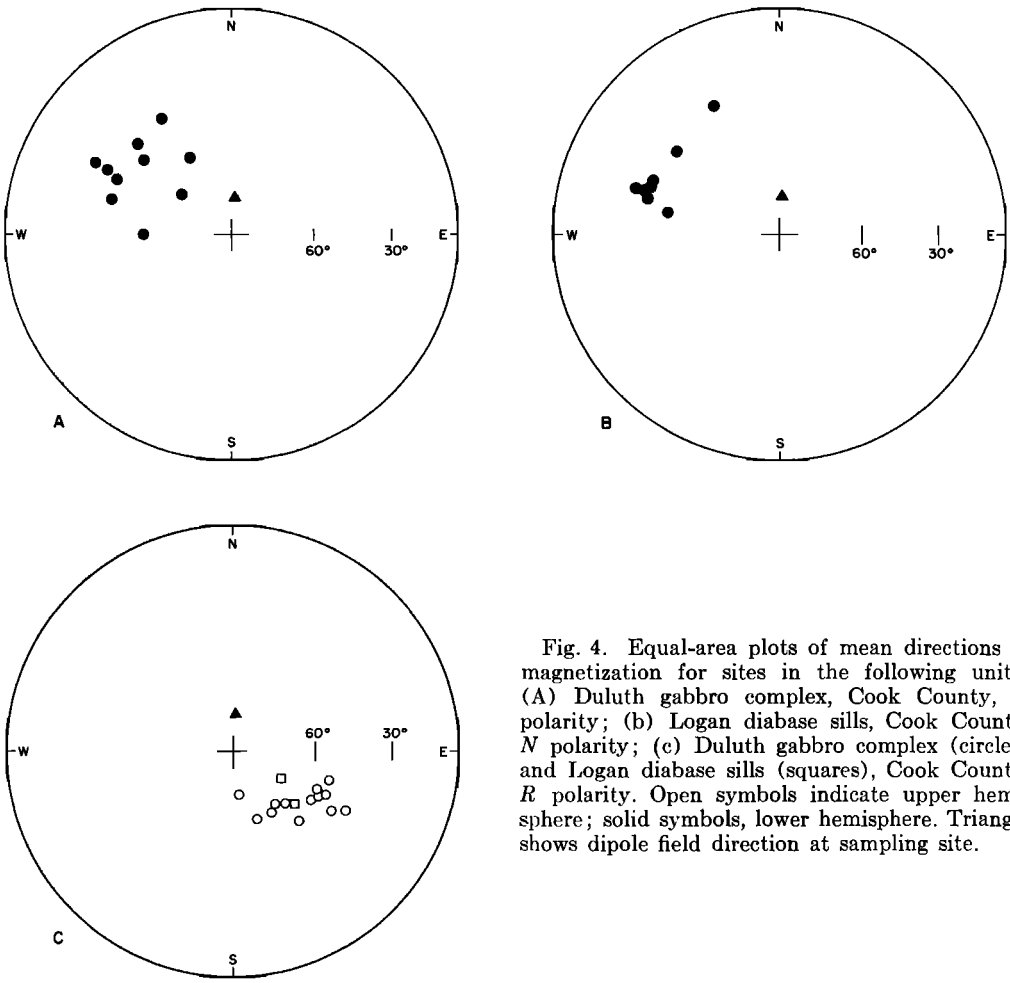


Fig. 4. Equal-area plots of mean directions of magnetization for sites in the following units: (A) Duluth gabbro complex, Cook County, *N* polarity; (b) Logan diabase sills, Cook County, *N* polarity; (c) Duluth gabbro complex (circles) and Logan diabase sills (squares), Cook County, *R* polarity. Open symbols indicate upper hemisphere; solid symbols, lower hemisphere. Triangle shows dipole field direction at sampling site.

in Duluth gabbro complex and Logan diabase sills are shown before tectonic corrections in Figure 4, and means for geological units corrected for a dip of 20°S are listed in Table 3. Only two stable sites in *R* Logan diabase sills are available and, as shown in Figure 4C, these have directions of magnetization that fall within the cluster of site-mean directions for the *R* gabbro sites. Data for the *R* diabase and gabbro sites have been combined in Table 3 because similarity of composition and magnetic directions and proximity of occurrence in the field indicate that the two units are of the same age.

Paleomagnetic data and locations of individual sites from all units listed in Tables 2 and 3 are given in *Beck* [1969, Appendix].

Magnetic intensities. Magnetic intensities of Minnesota Keweenawan intrusive rocks vary enormously, in keeping with the great variability shown in other lithological properties. Much of this variability is found within the individual intrusions themselves and apparently originates in response to processes of magmatic differentiation [*Beck*, 1969, pp. 97–116]. Intensities of natural remanent magnetization (NRM) in Minnesota Keweenawan intrusive rocks range from about 3 to 30,000 $\times$ 10^{-6} emu; magnetic susceptibility (K) varies over nearly as wide a range. As an example of variability within a single intrusive unit, the layered series of the Duluth gabbro complex shows the following range (in emu $\times$ 10^{-3}): 5000 > NRM > 1; 500 > K > 5. Geometrical mean values of NRM and K for various rock types are summarized in

Table 4. Distribution of NRM and K is approximately log-normal.

Discussion

Figure 5 shows that paleomagnetic poles for *N* polarity Keweenawan intrusive rocks from Minnesota form a tight cluster. The *R* pole from Cook County lies a few degrees outside the cluster, probably because it represents an earlier point on a generally southwestward-trending path of apparent polar wandering [*Beck and Lindsley*, 1969; *Beck*, 1969]. The same apparent southwestwardly drift of the paleomagnetic pole may also be recorded in the cluster of *N* poles. On geological grounds the layered series (L) is younger than the anorthositic gabbro (AN); the Cook County diabase (CLN) may be younger than the Cook County gabbro (CDN); and the Beaver Bay Complex, which is present high in the section of Keweenawan lavas and is undeformed, may be the youngest intrusion of the group [*Taylor*, 1964; *Grout et al.*, 1959; *Gehman*, 1957].

One pole (AN) in this group may be suspect. As shown in Figure 3A, the anorthositic gabbro has an elongate distribution of site-mean directions of magnetization that at first glance suggests streaking between a primary and a secondary magnetic direction. Secondary magnetization is unlikely to be the cause of apparent streaking in this instance, however, because (1) within-site streaking is not present in anorthositic gabbro sites after af demagnetization even in sites that consist of five or six samples of highly variable lithology; (2) no known source of secondary magnetization lies at either end of the elongate pattern of site-mean directions.

As discussed more fully elsewhere [*Beck*, 1969, pp. 31–32], causes of apparent streaking other than magnetic instability are known. These include uncorrected differences in amount of tectonic rotation about a common line of strike, differences in degree of shape or crystallographic anisotropy, and apparent polar wandering during the magnetization process. Neither differential rotation nor anisotropy is likely to apply to the anothositic gabbro; known tectonic trends are such as to produce an E–W direction of apparent streaking not to be confused with the observed direction, and anisotropy is extremely unlikely in this massive, coarse-grained rock. Very coarse grain size suggests extremely slow cooling, so perhaps enough time elapsed during magnetization of the anorthositic gabbro to record motion of the paleomagnetic pole. Alternatively the elongate pattern of Figure 3A could be simply an accidental effect of inadequate sampling. Whatever the source of apparent streaking, the anorthositic gabbro has been accepted as magnetically stable for the purposes of this report. Its distribution of directions is markedly non-Fisherian [*Fisher*, 1953], so that dispersion statistics based on Fisher's distribution (as given in Table 2) technically are inapplicable.

Average pole for Keweenawen intrusive rocks. It is useful to have an average paleomagnetic pole position for Minnesota Keweenawan intrusive rocks (including the Beaver Bay complex) with which to compare paleomagnetic pole positions for other geological units. Such

TABLE 3. Directions of Magnetization and Paleomagnetic Pole Positions for Rock Units from Cook County

Unit	No. Sites	D	I	k	α_{95}, deg	λ'	ϕ'	δP_{95}, deg	δm_{95}, deg
Duluth gabbro complex, *N* polarity	10	280.0	+56.0	25.2	10.0	32.5°N	160.0°W	10.5	14.5
Logan diabase intrusives, *N* polarity	8	277.5	+47.0	37.3	9.0	25.5°N	166.0°W	7.5	11.5
Duluth gabbro complex and Logan diabase intrusions, *R* polarity	13	91.0	−65.5	44.2	6.0	42.5°N	156.5°W	7.0	9.5

For explanation of column headings, see Table 2.

TABLE 4. Geometrical Mean of Intensity of Natural Remanent Magnetization (NRM) and Magnetic Susceptibility (K) for Several Types of Minnesota Keweenawan Intrusive Rocks

Rock Type	NRM, X 10^{-4} emu	K, X 10^{-4} emu
Anorthosite	2.8	1.3
Anorthositic gabbro	10.7	7.3
Diabase	3.2	2.4
Ferrogranodiorite	9.7	24.4
Granophyre	3.4	13.6
Oranophyric diabase	12.3	21.2
Layered gabbro	7.4	4.7
Melagabbro	4.3	4.5
Pegmatitic gabbro	55.0	25.2

an average pole position can be computed using as data the unit-poles of Tables 2 and 3 and the Beaver Bay complex pole of Table 1, or it can be based on poles for the individual mean directions for each site. Neither practice seems clearly superior in this case. Results of calculations by both methods are given in Table 5. Reversely magnetized sites from Cook County are excluded because they may be significantly older than sites from the *N* units. Calculation 1 included, in addition to the raw data of calculation 2, the 11 melagrabbro, ferrogranodiorite, and granophyre sites of Figure 3C. Also included in calculation 1 are three sites from the Duluth area located on the contact between units of the Duluth gabbro complex and consequently not included in paleomagnetic pole calculations for any one unit. In this manner data are incorporated in calculation 1 that could not be used in calculation 2, making calculation 1 the preferred average pole position for *N* polarity Keweenawan intrusive rocks from Minnesota.

The fact that dispersion of the 7 unit poles is much less than dispersion of the 131 site poles (k' of calculation 1 less than k' of calculation 2) probably indicates that the separate gabbro units in part cooled during the same interval of time, thus 'sampling' an identical distribution of geomagnetic field directions. Thus the dispersion of calculation 1 might be considered analogous to the variance of a sample drawn from a normal population, whereas the dispersion of calculation 2 more nearly represents the variance of the sampling distribution of the mean. Dispersion and confidence figures cited in Table 5 should be regarded as overestimates because they include an unknown contribution from small undetected differences in true geological tilt from site to site. The actual amount of tilt experienced at a particular site could be determined only in rare instances because of the scarcity of exposures of bedded rock. The figures used (20°S and 20°E) are rough averages of data from published geological maps and from the author's observations. Deviations from these averages are unlikely to be significant, and the dispersion and confidence-limit figures of Table 5 can be regarded as nearly correct.

Secular variation. *Creer* [1962] and *Cox* [1962] have analyzed the dispersion in magnetic directions that should result at various points on the earth's surface from the observed westward drift of the current (Epoch 1945.0) nondipole field. Results were found to be strongly latitude dependent and to show marked asymmetry between northern and southern hemisphere and between the Pacific region and the rest of the world. *Creer* [1962, Figure 10] further demonstrates a tendency for the modern latitude dependence to have persisted since the Permian. If the Creer-Cox model of secular variation is applicable, dispersion in directions of remanent magnetization can be corrected ap-

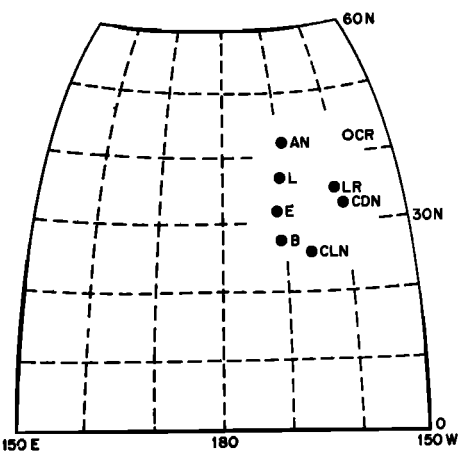


Fig. 5. Equal-area plot of poles from Tables 2 and 3. Pole B, Beaver Bay Complex, from *Beck and Lindsley* [1969]. AN indicates anorthositic gabbro; L, layered series; E, Endion sill; LR, Lester River sill; CDN, Duluth gabbro complex, Cook County, *N* polarity; CLN, Logan diabase sills, Cook County, *N* polarity; CR, Duluth gabbro complex and Logan diabase sills, Cook County, *R* polarity.

TABLE 5. Average Paleomagnetic Pole Position for N-Polarity Intrusive Rocks Units of Keweenawan Age from Minnesota

Calculation	No. Sites	λ'	ϕ'	k'	α_{95}, deg
1	131	34.5°N	168.0°W	28.9	2.2
2	7	33.0°N	167.0°W	143.3	5.1

Calculation 1 uses site-mean poles as data. Calculation 2 uses poles for individual geological units as data. λ', ϕ', and α_{95} as explained in Table 2; k' is Fischer's dispersion parameter for paleomagnetic poles. For the paleolatitude of these intrusions, $k \approx k'$ [*Creer*, 1962, Fig. 7].

proximately for the effect of differences in paleolatitude. In this way it is possible to estimate the value of the ratio of nondipole to dipole field intensity for various points in geological time, relative to the same ratio for the modern geomagnetic field. An interesting recent suggestion by *Cox* [1968] links high frequency of geomagnetic field reversal to high ratios of nondipole to dipole field intensity.

Paleolatitude for the Minnesota Keweenawan intrusive rocks is about 35°. From Figure 10 of *Creer* [1962], rock units becoming magnetized at a latitude of 35° today might be expected to have a dispersion of about 14° in terms of angular dispersion factor, $\cos^{-1} R/N$ [*Wilson*, 1959]. From Creer's compilation of Permian and later results, dispersion of 9°–19° might be expected. Results for seven N-polarity Keweenawan intrusive units range from 7.5° to 15.5°. Evidently on this model the ratio of nondipole to dipole field intensity 1 billion years ago was not markedly different from its average value since Permian time, although it

TABLE 6. Paleomagnetic Pole Positions for some Mafic Rock Units of Precambrian Age from North America

Symbol on Figure 6	Unit	λ'	ϕ'	Polarity	Reference
A	Abitibi dikes	27°N	134°W	N	*Larochelle* [1966]
CH	Basalts in Copper Harbor conglomerate	28°N	178°W	N	Recalculated from *DuBois* [1962]
CM	Lavas at Coppermine	1°N	176°W	N	*Robertson* [1964]
CR	Cook County diabase and gabbro, reverse polarity	34.5°N	141.5°W	R	This paper
G	Flows near Grand Portage, Minn.	44°N	163°W	R	Calculated from *Books* [1968]
I	Flows near Ironwood, Mich.	29°N	128°W	R	Calculated from *Books* [1968]
L	'volcanics and sills from Lake Superior area'	50°N	148°W	R	*Palmer* [1968]
MA	Marathon dikes	29°N	147°W	N	*Fahrig et al.* [1965]
ME	Gabbro near Mellen, Wisconsin	36°N	175°W	N	*Books et al.* [1966]
MI	Minnesota Keweenawan intrusives, N polarity	34.5°N	168°W	N	This paper
MO	Molson dikes	36°N	109°W	N	*Fahrig et al.* [1965]
MK	Mackenzie dikes	4°N	177.5°W	N	*Fahrig et al.* [1965]
MU	Muskox intrusion	4°N	175°W	N	*Robertson* [1964]
P	Portage Lake lava series	27°N	179°W	N	Calculated from *Books* [1968]
SA	Diabase at Sierra Ancha	27.5°N	178°W	...	*Helsley* [1965]
SD	Sudbury dikes	2°S	171°W	N	*Larochelle* [1967]
SI	Sudbury intrusion	47°N	107°W	N	*Sopher* [1963]
1.34	Baraga County, Mich.; dikes	45°N	99.5°W	R	*Graham* [1953]
1.66	Purcell lavas	29°S	144°W	R	*Black* [1963]
1.80	Lavas at Mamainse Point	31.5°N	173.5°W	N	*DuBois* [1962]
1.81	Logan diabase, I	53.5°N	130°W	R	*DuBois* [1962]
1.82	Logan diabase, II	33°N	171.5°W	R	*DuBois* [1962]
1.84	Lavas at Alona Bay	39°N	95°W	R	*DuBois* [1962]

was a little smaller than at present. If the nondipole field originates in processes taking place at or near the surface of the earth's liquid core, whereas the dipole field arises from volume-related processes, the ratio of nondipole to dipole field intensity might be expected to decrease with increase in the radius of the core. In this way the dispersion of paleomagnetic directions in Keweenawan intrusive rocks argues against marked growth of the core since Keweenawan time.

Paleomagnetism. Table 6 and Figure 6 give paleomagnetic pole positions for some mafic rock units of Precambrian age from North America. This is not a complete compilation. Only the latest work is shown in cases where duplicate poles are available. For instance, pole MI from this report is plotted instead of DuBois's Duluth gabbro complex pole [*DuBois*, 1962] and the Beaver Bay Complex pole of *Beck and Lindsley* [1969]. In similar manner the Abitibi dikes pole ENEW of *Larochelle* [1966] replaces the Abitibi dikes pole of *Fahrig et al.* [1965], and the Portage Lake lavas pole of *Books* [1968] replaces poles for the same unit by *DuBois* [1962] and *Vincenz* [1968]. Other poles arbitrarily ignored include the Matachewan dike poles of *Fahrig et al.* [1965], eliminated because of their great (and uncertain) age relative to the other poles of Figure 6, and the Abitibi dike ENEW and NNE poles of *Larochelle* [1966], eliminated because they probably are based on too few localities to be significant.

Despite its limitations, Figure 6 suggests several important interpretations.

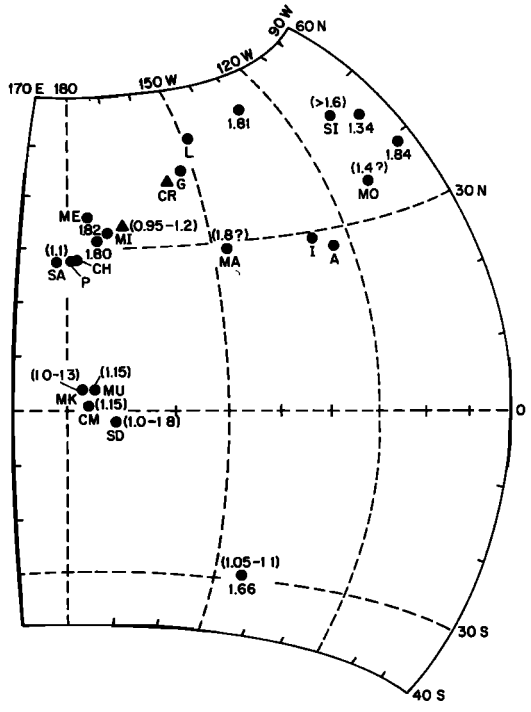


Fig. 6. Equal-area plot of paleomagnetic pole positions for some Precambrian mafic rock units from North America. Dots indicate previously published poles summarized in Table 6; triangles, new poles published in this paper. Numbers within parentheses are radiometric ages, in billions of years, as reported by the author of the paleomagnetic study from which the pole was taken. Numbers not in parentheses refer to Irving's compilation of paleomagnetic poles [*Irving*, 1964].

1. Basaltic igneous activity seems to have occurred sporadically throughout much of the last 1.0 and 1.5 b.y. of Precambrian time in North America.

2. The practice of depicting the course of apparent polar wandering by use of a smooth *curve* is inappropriate for the more ancient divisions of geological time. Even for such an intensively studied subject as the late Precambrian of North America the course of apparent polar wandering is less realistically depicted as a line than as a *path* some few tens of degrees wide. Within the path the trace of the pole is still undecipherable, except in a few instances. Adding paleomagnetic poles for other Precambrian units to Figure 6 would not change this conclusion, especially since lack of precision in radiometric dating probably is responsible for much of the apparent scatter [*Beck*, 1969, pp. 133–140].

3. As pointed out by *Larochelle* [1966] and emphasized by Figure 6 of this report, the recurrent nature of basaltic igneous activity in the Canadian Shield (and elsewhere in North America) makes long-distance correlation by paleomagnetic techniques somewhat hazardous. Figure 6 shows that such techniques can nevertheless be useful in certain instances. For example, the Sierra Ancha dike from central Arizona (SA of Figure 6; *Helsley*, 1965) must correlate with Keweenawan activity in the Lake Superior area rather than with the basaltic activity represented by the Muskox intrusion, Coppermine lavas and Mackenzie dikes (poles

MU, CM, MK). This can be concluded from paleomagnetic evidence even though the Keweenawan and Mackenzie area poles represent rock units with very similar ranges in radiometric age. Conversely, the Sudbury dikes (SD) must correlate with the Mackenzie rather than with the Keweenawan rock units [*Larochelle*, 1967; *Fahrig and Jones*, 1969], even though the latter are geographically much closer. Furthermore, the lower limit of the reported range in possible age for the Sudbury dikes (1.0-1.8 b.y.) appears to be the more likely. Finally, the Purcell Basalts (1.66) and diabase sill (not shown on Figure 6) studied by *Black* [1963] appear to correlate with neither the Keweenawan nor Mackenzie igneous events, even though the radiometric age obtained by *Hunt* [1962] places the Purcell activity within the time span recorded for both Keweenawan and Mackenzie events. Evidently paleomagnetism can be useful as a supplement to other long-distance correlation techniques in certain favorable instances.

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